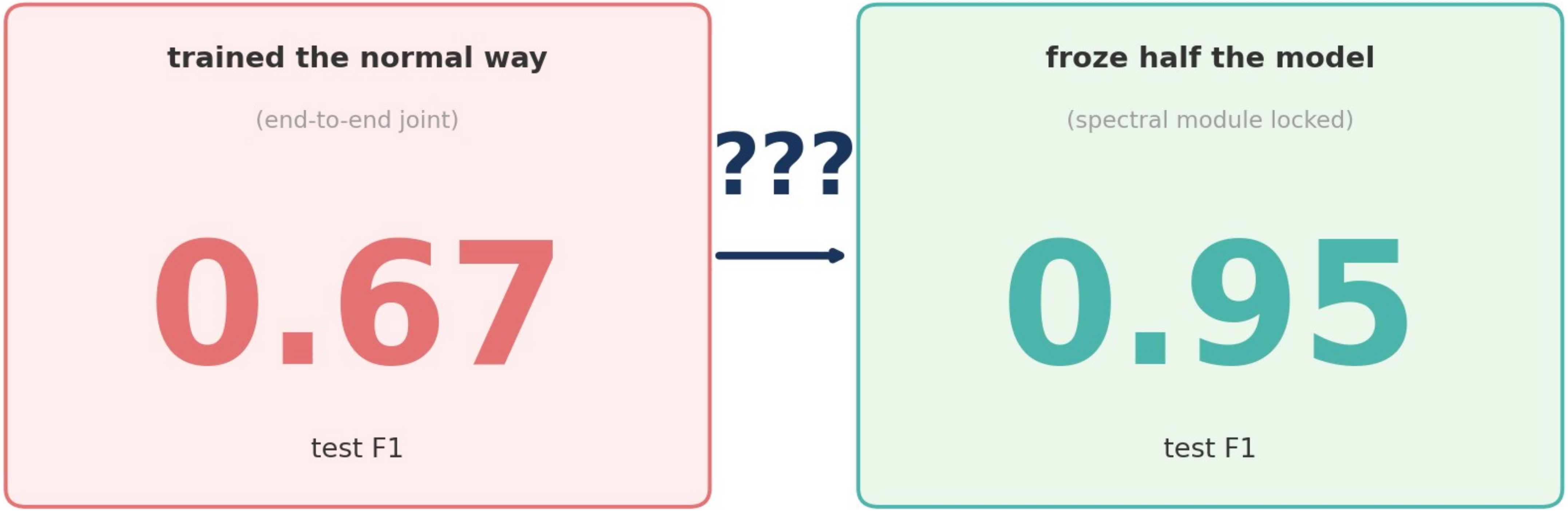
The Spectral Shortcut

why locking part of a model makes it better

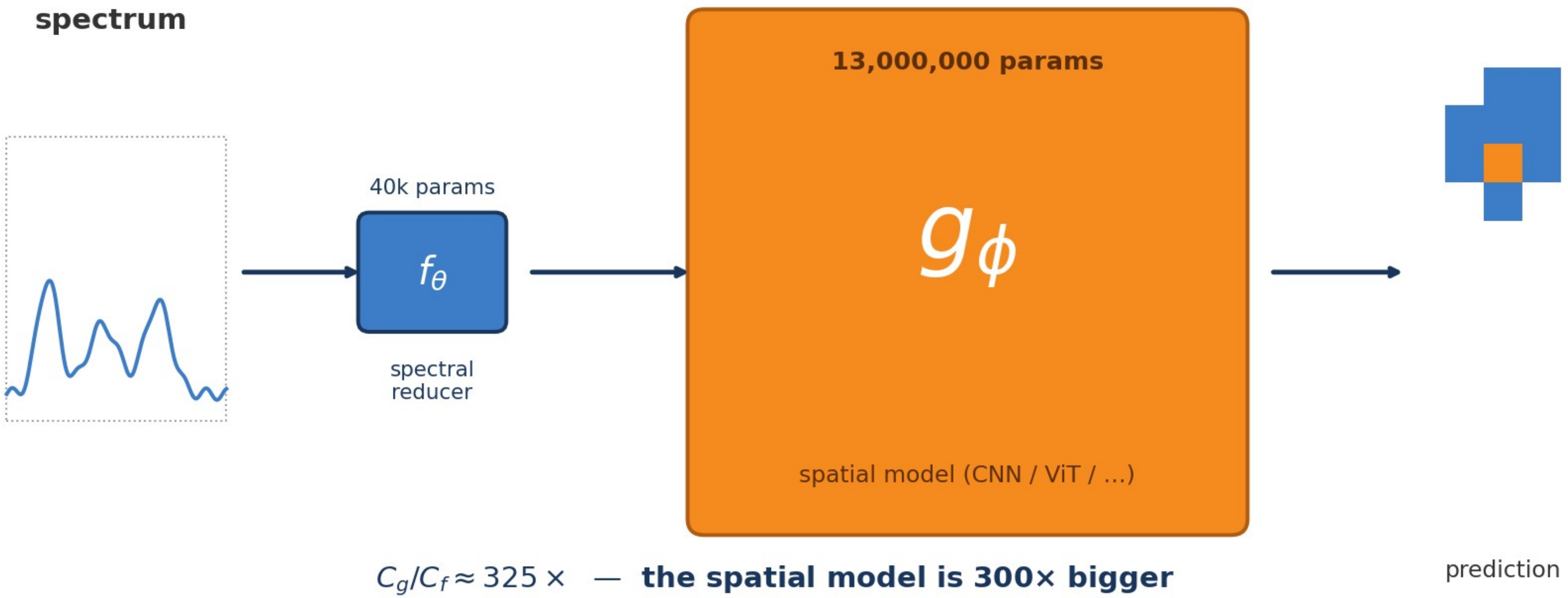
Krzysztof Dziuba · lab meeting

the puzzle



same model — totally different results

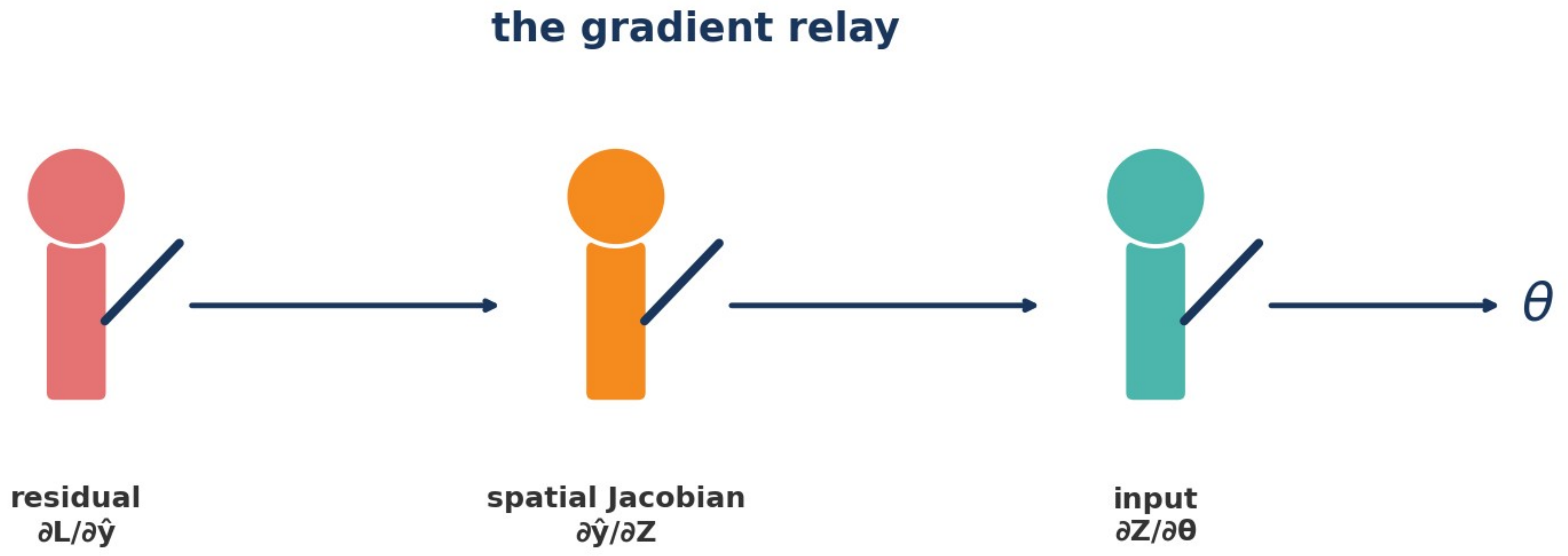
what's inside



a small "squeezer" feeds a huge "brain" — and the size gap is the whole story

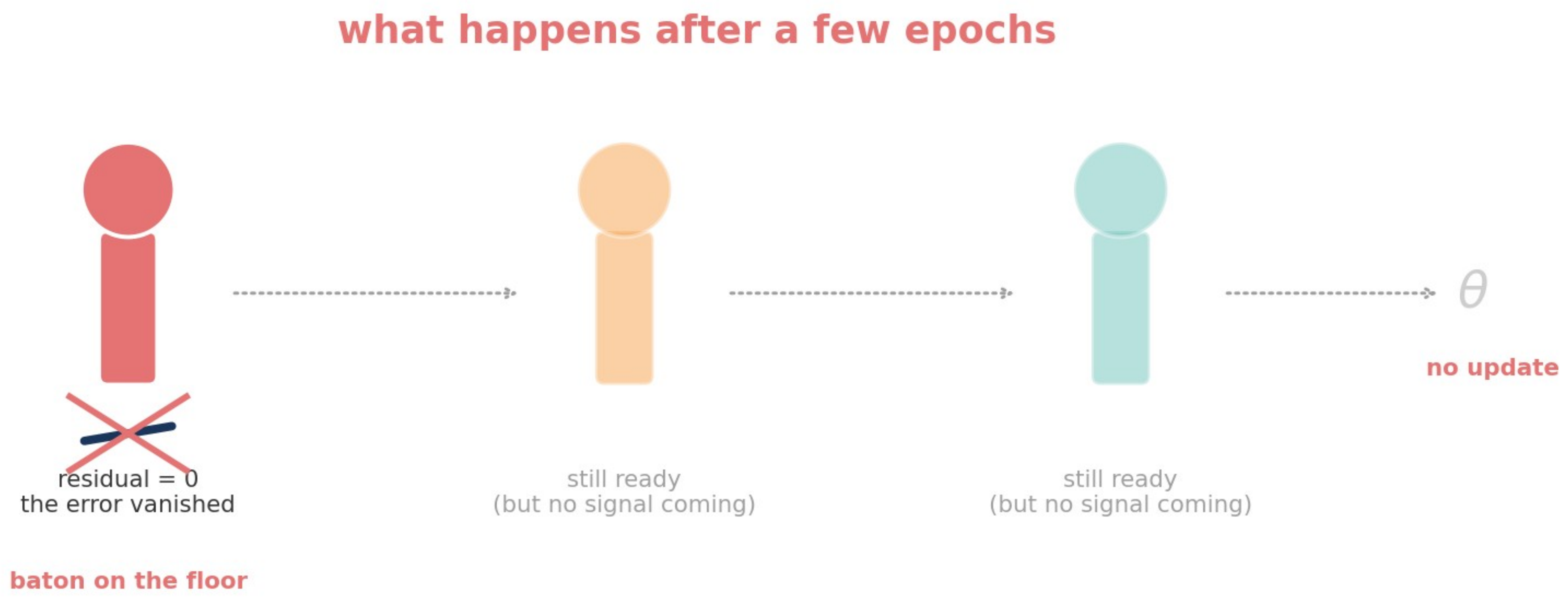
gradients move backward — like a relay

$$\partial L/\partial \theta = \partial L/\partial \hat{y} \cdot \partial \hat{y}/\partial Z \cdot \partial Z/\partial \theta$$



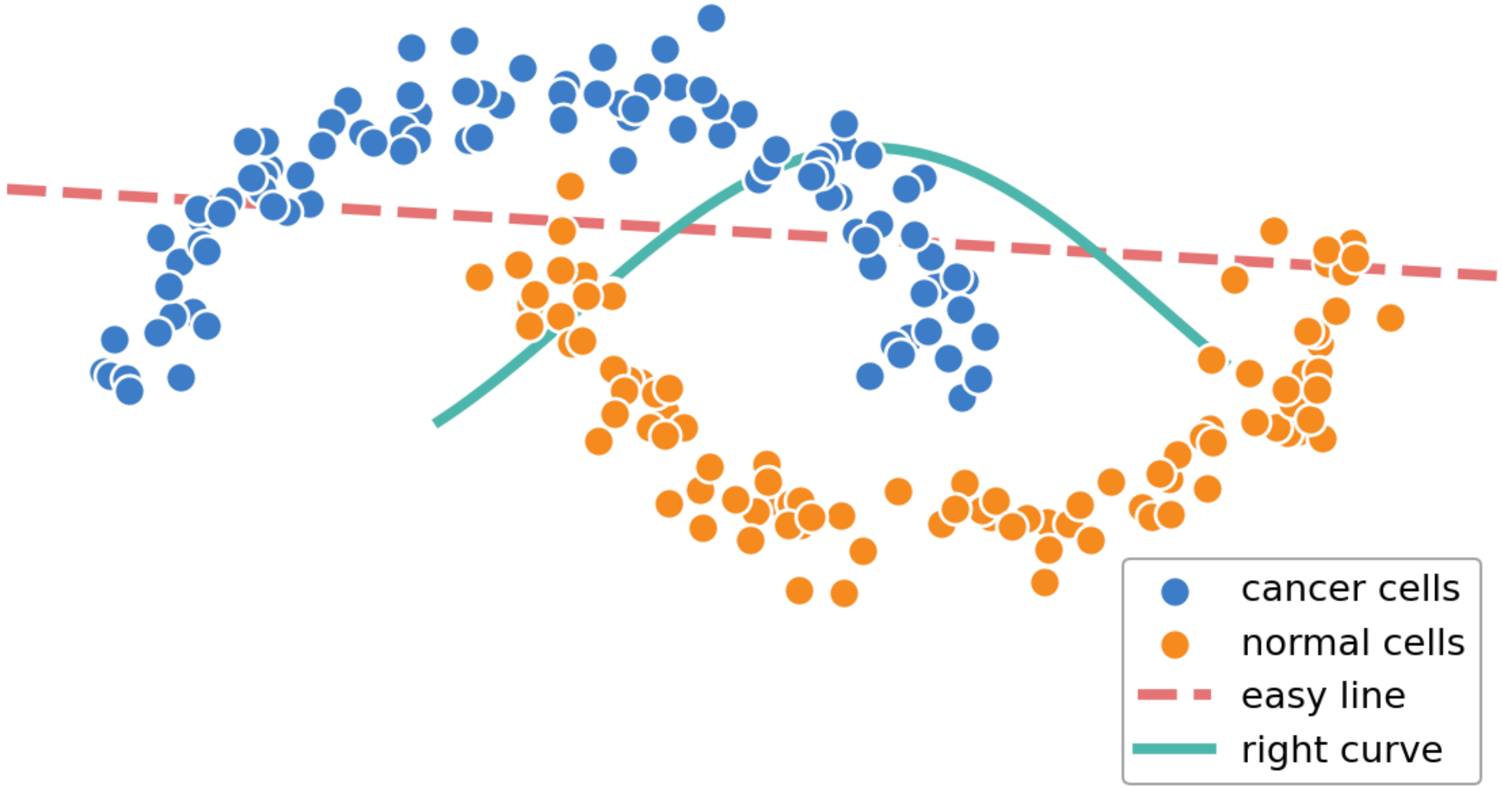
three runners pass the same baton: "residual" → "Jacobian" → "input" → θ

and then the first runner drops it



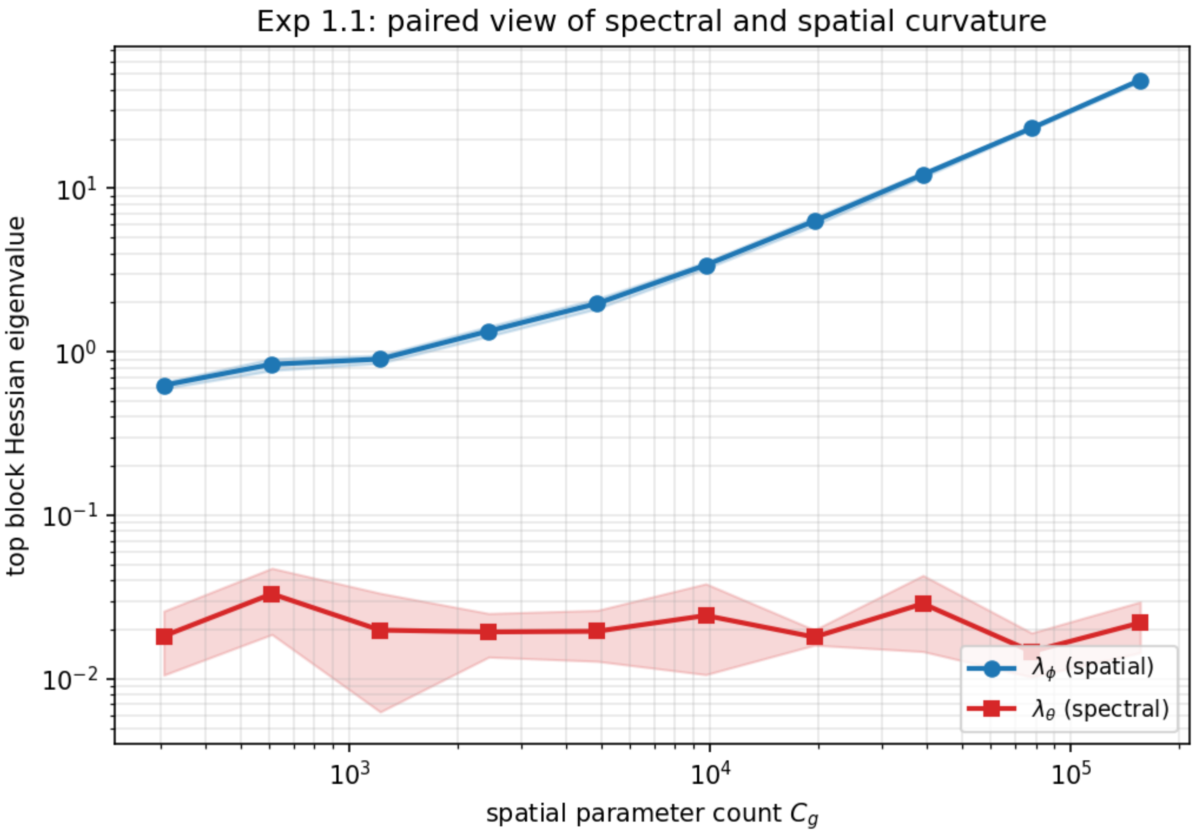
cross-entropy saturates fast → residual = 0 → no gradient ever reaches θ

the "easy line" beats the "right curve"



imagine cancer (blue) vs normal (orange) — gradient descent picks the easy line, ignores the curve. In our case the "easy line" is morphology, the "right curve" is chemistry.

and the data shows it



spatial curvature (blue) grows 75×.
spectral curvature (red) stays flat.

freeze it

lock the spectral module outright

or anchor it

inject a fixed baseline (PCA, etc)

watch the meter

track EGR during training as a signal

freeze. anchor. watch.

don't train everything at once — the math says you can't.